the percentage of system resources that a soundcard consumes while it processes and outputs sound at 16 bit and 22 KHz frequency.

c) Technical test: RMAA (version 3.4) is an exhaustive test for a soundcard. This suite runs five subtests, namely, Frequency response, Signal-to-Noise level, Dynamic range, THD + Noise and Stereo Cross talk.

Warranty and support: Here we took into account the warranty period for the soundcards, the number of authorised service centres appointed by the companies and the number of cities in which authorised service centres are present. The information here is as provided by the respective companies.

Value for money: This provides us with a price-performance ratio. A card offering good performance at a great price will score higher than one offering excellent performance with a hefty price tag.

How they fared

We tested all the major soundcards that are available in the market today. Products from Creative Labs dominated our test line-up with five out of the six cards coming from their stables; the other card was provided by Zoltrix. We also tested two integrated sound chipsets—these reside on the motherboard and are an ideal solution for those on a budget.

Features

The number of output channels supported by a soundcard is an extremely important feature to keep in mind while making a purchase, as this cannot be upgraded or modified. The price of 2.1 and 4.1 speaker sets has come down drastically and games and movies are a joy to experience on these multi-speaker sets. The Creative Vibra 128 was the only soundcard that did not support more than a 2.1 speaker set, while the remaining supported either 4.1 or 5.1 speakers. Older cards had a built-in amplifier that created distortion and caused loss in the purity and clarity of the sound output (this was a necessary evil as most speaker sets those days were non-amplified). All newer soundcards provide a non-amplified, pure output to the speakers with only the Vibra 128 still amplifying its line-out and MIDI jacks.

For a theatre-like audio experience, a soundcard should support Digital Dolby 5.1 decoding. Unfortunately, this feature is only available in Creative's Audigy Platinum, Sound Blaster Live! DE 5.1 and Extigy solutions.

Although not an issue for a majority of users, the nature of drivers available for those using niche operating systems such as Linux and Windows 3.1 can be a point of concern. Without the proper drivers, users will get sound of questionable quality.

DID YOU KNOW

Solutions for the Musician

Some cards only have FM synthesizers and these should be considered a bad choice to create quality music. A musician should buy a soundcard that features Wavetable support and optionally supports the SoundFont technology (see *www.soundfont.com* for more information). Finally, if you wish to reduce signal loss, make sure that your soundcard has two things: gold-plated connectors and an S/PDIF connection port, along with the requisite cables.

Jargon Buster

Duplex: A full-duplex soundcard can make and receive sounds at the same time. A full-duplex soundcard is thus essential if you intend to use telephony over the Internet. For instance, walkie-talkies are half-duplex, while telephones are full-duplex.

MIDI Channels: These channels offer a musician greater control over the instruments connected to the soundcard. He can set them all to respond to different channels and therefore have independent control over each one.

Polyphony: Is the maximum number of voices a synthesiser can play at any one time. A soundcard functions around a Digital Signal Processor (DSP) that mixes sound samples from the on-board wavetable to produce the requisite musical notes. The maximum number of notes available is dependent on the processing power available in the DSP and is referred to as the card's 'polyphony'.

Recording/Playback Depth and Rate: In order to store sound in digital form, its waveform is converted into a series of bits via a process called sampling. A higher sampling rate will give a more accurate digital representation of the sound. The maximum sampling rate at which a sound signal can be recorded by a soundcard is called the recording rate of the card; the higher this value, the better the recording. Similarly, a higher playback rate will give better sound quality. The clarity and fidelity of the sound sample improves with the number of bits used to describe it. Thus, a 16-bit sample recorded at 24 KHz sampling rate will have less clarity than a 24-bit sound sample recorded at a rate of 96 KHz.

Signal-to-Noise Ratio (SNR): SNR is the ratio of the largest sound signal that can be handled by a card with minimum distortion, to the noise that is present at that time. It thus tells us how 'cleanly' the card in question produces sound.

Sony/Philips Digital Interface (S/PDIF): It is a standard for transmitting data in a lossless digital format to preserve sound quality. Soundcards that feature an S/PDIF connector are thus a better choice.

Stereo Cross talk: Cross talk is the mixing of the left and the right-channel sound information. The effect being, when listening to a set of stereo speakers (or headphones), the left ear can hear sounds meant for the right speaker, and the right ear can hear some of the audio coming from the left speaker. A well-engineered soundcard (one with gold-plated connectors and shielded wires, for instance) will suppress this interference and produce a more accurate sound.

Total Harmonic Distortion (THD): Non-linear distortion is a processing error that creates output signals at frequencies that are not necessarily present in the input. THD is determined by measuring the size of each of the new frequencies that are created by the source of the distortion and then taking their geometric sum.

Wavetable: A Wavetable stores the digitised samples of actual recorded instruments, which are then combined during music creation and playback. They are commonly used by professionals to synthesise music.